

1 Critical evaluation of the modified Gompertz response
2 function as a parametric template for
3 treatment-response trajectories in N-of-1 trial
4 simulation

5 pmsimstats team

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35 1 Abstract

36 **Background.** Simulation-based power analyses for aggregated N-of-1 trials re-
 37 quire a parametric template for the within- participant response trajectory. The
 38 pmsimstats framework adopts a modified Gompertz function for each of three
 39 latent response components (biological response, placebo-belief response, time-
 40 variant natural-history component). The choice of family is loaded into the simu-
 41 lated data before any analysis-model question is asked, so the implications of the
 42 family for power, type I error, and bias are foundational rather than peripheral.

43 **Methods.** We evaluate the modified Gompertz family against the relevant biolog-
 44 ical, pharmacokinetic, and biostatistical literature. We characterise the parameter
 45 space the function covers, the trajectories it can and cannot represent, and the
 46 sensitivity of standard pmsimstats inference to misspecification of the response-
 47 curve family. Three alternative families are compared as sensitivity analyses:
 48 the logistic (symmetric sigmoidal), the hyperbolic-tangent (fast-saturating), and
 49 a piecewise-linear breakpoint model (no smoothness assumption). For each al-
 50 ternative we compute power, type I error, and bias of the biomarker-treatment
 51 interaction estimate at trial-relevant sample sizes against simulated data drawn
 52 from the alternative family but analysed under the standard linear-mixed model.

53 **Results.** A 1{,}000-replicate-per-cell simulation across 16 cells (4 trajectory fami-
 54 lies $\times$ 2 DGP architectures $\times$ 2 biomarker effect-size levels) at $N = 35$, OL+BDC
 55 design, $t_{1/2} = 0$, found trajectory family essentially irrelevant to power under
 56 both architectures at MCSE 0.013 to 0.016. Under Architecture A (mean moder-
 57 ation), the four families produced powers in $\{0.761, 0.769, 0.771, 0.778\}$, a 0.017
 58 spread within 2 MCSE of sampling noise. Under Architecture B (MVN differen-
 59 tial correlation), powers clustered at $\{0.779, 0.780, 0.781, 0.806\}$; logistic at 0.806
 60 sat $\approx$ 2 MCSE above the others, marginal and not reproduced under Architecture
 61 A. The piecewise-linear breakpoint family, hypothesised to behave qualitatively
 62 differently from the smooth families, was indistinguishable from gompertz and
 63 hyperbolic-tangent at this resolution under either architecture. Type I error at
 64 the null sat in $[0.025, 0.036]$ across all eight null cells, slightly conservative and
 65 without family-specific inflation.

66 **Conclusions.** The modified Gompertz family is a defensible default for the
 67 response-curve template in pmsimstats N-of-1 simulation, with explicit caveats for
 68 the trajectory regimes it cannot represent. We recommend reporting Gompertz-

69 vs-alternative sensitivity at least once per parameter regime where simulation
70 results inform protocol-design decisions, and we propose a default sensitivity grid
71 that the framework can run alongside its primary cells.

72 2 Introduction

73 2.1 A claim about response-curve choice in N-of-1 trial simulation

74 **The Gompertz function is a defensible default for the saturating-shape**
75 **regime in N-of-1 simulation.**

- 76 • Its asymmetric three-parameter form spans profiles that logistic and expo-
77 nential families cannot represent.
- 78 • Established precedent in tumour dynamics, microbial growth, and the
79 unified-Richards family supports the choice.
- 80 • The claim is conditional on the saturating-shape regime characteristic of
81 most pharmacodynamic trajectories over typical trial timescales.

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89 *The first claim is that the modified Gompertz function used as the para-*
90 *metric template for treatment-response trajectories in aggregated N-of-1 trial*
91 *simulation (specifically, in the pmsimstats simulation framework that oper-*
92 *ationalises the Hendrickson et al. design family [hendrickson2020]) is a*
93 *defensible default for the saturating-shape regime that characterises most*
94 *pharmacodynamic and chronic-condition response curves over the typical*
95 *trial timescale. The function is well-established in tumour growth dynamics*
96 *[laird1964dynamics; norton1988gompertzian], microbial growth modelling*
97 *[zwietering1990modeling], and the broader unified- Richards growth-curve*
98 *family [richards1959flexible; tjorve2010richards; tjorve2017]; its three-*
99 *parameter form is flexible enough to capture the asymmetric saturation pro-*
100 *files that linear extrapolation, exponential decay, or symmetric logistic models*
101 *cannot.*

102 **Family-misspecification sensitivity is absent from the N-of-1 literature and**
103 **should be reported routinely.**

- 104 • Existing N-of-1 simulation studies inherit trajectory families without com-
105 paring alternatives.
- 106 • Oncology and pharmacology modelling have established comparative appa-
107 ratus that the N-of-1 field has not imported.
- 108 • Routine sensitivity reporting is needed rather than silent assumption of
109 family adequacy.

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117 *The second claim is that family-misspecification sensitivity is essentially ab-*
118 *sent from the published N-of-1 methodology literature, and that it should be*
119 *reported routinely rather than assumed away. The methodological N-of-1 cor-*
120 *pus [@lillie2011; @schork2023precision; @vohra2015cent; @hendrickson2020]*
121 *is, to the best of our knowledge, silent on parametric trajectory shape: papers*
122 *either inherit a chosen family from the upstream literature without comparing*
123 *alternatives, or specify a linear or AR(1) form that is too coarse to capture*
124 *the saturation that real trajectories exhibit. The oncology-modelling literature*
125 *has set the precedent for what a proper family-comparison sensitivity analysis*
126 *looks like [@benzekry2014classical; @tjorve2010richards], and the pharma-*
127 *cology literature has converged on Hill / sigmoid-Emax / indirect-response*
128 *models for related questions [@goutelle2008hill; @dayneka1993indirect]. The*
129 *N-of-1 literature has not imported this comparative apparatus, and the present*
130 *paper endeavors to supply it.*

131 **The case for Gompertz is incomplete without a direct comparison of**
132 **inferential consequences across alternative families.**

- 133 • The paper does not assert Gompertz is uniquely correct but argues it is
134 reasonable within the unified-Richards class.
- 135 • Inferential consequences (power, type I error, bias) under alternative fami-
136 lies must be quantified, not asserted.
- 137 • The simulation study uses the standard pmsimstats linear-mixed analysis
138 at trial-relevant sample sizes.

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146 *The two claims are coupled because the case for a chosen family is incomplete*
147 *without the comparison. We do not argue that Gompertz is the correct family*
148 *in any strict sense; rather, we argue that it is a reasonable choice within*
149 *the unified-Richards family of saturating curves, and that the inferential con-*
150 *sequences of choosing it (versus the symmetric logistic, the von Bertalanffy*
151 *growth curve [@vonbertalanffy1957], or a piecewise-linear breakpoint model)*
152 *should be quantified rather than asserted. The paper reports a simulation*
153 *study that does so for the standard pmsimstats linear-mixed analysis at trial-*
154 *relevant sample sizes.*

155 2.2 Why the question matters

156 **Family choice is a DGP-level decision whose inferential cost cannot be**
157 **detected without varying the DGP family while holding the analysis fixed.**

- 158 • A single-family simulation cannot reveal the inferential cost of that family
159 choice.
- 160 • Robustness to family is a joint property of the DGP and analysis specifica-
161 tion.
- 162 • This is analysis-model misspecification logic applied one level down in the
163 simulation hierarchy.

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171 *First, the chosen family loads into the data before any analysis-side question*
172 *is asked. A power simulation that draws every trajectory from a Gompertz*
173 *curve and then analyses each under a standard linear mixed-effects model*
174 *with treatment indicator and AR(1) residual correlation cannot detect the*
175 *inferential cost of the family choice itself. Robustness of the inference to*
176 *the family is a property of the joint DGP-and-analysis specification; only a*
177 *comparative simulation that varies the DGP family while holding the analysis*
178 *fixed can quantify it. This is the same logic that underpins analysis-model*
179 *misspecification studies in the broader clinical-trial methodology literature*
180 *[@morris2019], applied one level down in the simulation hierarchy, and it*
181 *appears to us that the N-of-1 simulation literature has not yet drawn the*
182 *parallel explicitly.*

183 **The Gompertz parameterisation has known identifiability fragility in short**
184 **noisy trajectories that has not been characterised for N-of-1 designs.**

- 185 • Asymptote and rate parameters correlate strongly; the shape parameter is
186 near-unidentifiable with few inflection- region timepoints.
- 187 • Unified-Richards reparameterisations and sloppiness diagnostics exist but
188 have not been applied to N-of-1 trajectory data.
- 189 • Comparative identifiability results from tumour-growth modelling support
190 the need for an analogous N-of-1 diagnosis.

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198 *Second, the function family has known identifiability fragility in short noisy*
199 *trajectories. Joint estimation of the three Gompertz parameters (m, d, r)*
200 *in $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and*

rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region. @tjorve2010richards and @tjorve2017 propose unified-Richards reparameterisations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote) with lower correlation, and @jagadeesan2023sloppiness develops the formal 'sloppiness' diagnostic that quantifies this directly as a property of the parameter space. @benzekry2014classical fitted eight tumour-growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalised logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories, and its absence is, in our view, problematic.

The Gompertz default in N-of-1 simulation reflects historical borrowing from growth biology, not empirical comparison with pharmacology alternatives.

- The pmsimstats Gompertz form traces to tumour dynamics and microbial growth literature, not pharmacodynamic evidence.
- The pharmacology literature defaults to Hill/sigmoid-Emax and indirect-response models for comparable questions.
- The two traditions have not been compared; inheritance is not itself evidence of adequacy.

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Third, the default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via the broader growth-curve tradition, where its use traces to Laird's tumour dynamics work [@laird1964dynamics] and to Zwietering's microbial growth reformulations [@zwietering1990modeling]. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family [@hill1910; @goutelle2008hill] and to Jusko's indirect-response models [@dayneka1993indirect]. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defence against the pharmacology default. Either default may be adequate for the saturation regime, but the inheritance is not itself evidence, and the question merits a direct empirical answer.

2.3 What the published N-of-1 literature does and does not do

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No published N-of-1 simulation study has examined parametric trajectory-family choice or reported family-misspecification sensitivity.

- CONSORT-N-of-1, foundational programmatic statements, and the Hendrickson framework all omit trajectory-family comparison.
- Silent inheritance of the Gompertz form is pervasive across applied N-of-1 simulation work.
- Adjacent literatures (oncology, pharmacometrics, latent- class growth) have comparative apparatus that has not been applied to N-of-1 simulation.

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A review of the recent N-of-1 methodology literature confirms, to the best of our knowledge, a silence on the parametric-trajectory-family question.

Each major N-of-1 methodological strand silently inherits the Gompertz default without family comparison.

- CONSORT-N-of-1 reporting standards, Lillie and Schork programmatic statements, and the Hendrickson framework all omit family comparison.
- Applied prazosin-PTSD work inherits the framework without revisiting the trajectory assumption.
- No published N-of-1 simulation study, to the authors' knowledge, has broached the misspecification question.

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The CONSORT extension for N-of-1 trials [vohra2015cent] codifies reporting standards but does not address parametric trajectory shape. The foundational programmatic statements [lillie2011; schork2023precision] emphasise design and analysis broadly without committing to a parametric response form. The Hendrickson et al. framework [hendrickson2020] introduces the modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers in chronic-condition N-of-1 trials (including the prazosin-PTSD reference work building on raskind2013prazosin) inherit the framework without examining the choice. Unfortunately, this pattern of silent inheritance is pervasive enough that no published N-of-1 simulation study, to the best of our knowledge, has broached the family-misspecification question.

291 **Comparative trajectory-family apparatus exists in adjacent literatures but**
292 **has not been applied to N-of-1 simulation.**

- 293 • Oncology modelling, pharmacometrics, and latent-class growth literature
294 each have established multi-family comparison methods.
- 295 • None of these methods has been applied with the Gompertz form as the
296 N-of-1 simulation reference.
- 297 • Importing this machinery is the explicit contribution of the present paper.

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305 *In adjacent literatures the comparative apparatus does exist. The oncology-*
306 *modelling community routinely fits multiple growth families and compares*
307 *them on identifiability and predictive performance [benzekry2014classical].*
308 *The pharmacometrics community produces explicit sensitivity studies*
309 *for sigmoid-Emax and indirect-response models [ebeling1996feasibility;*
310 *@triggs1996levodopa]. The latent-class growth-trajectory literature*
311 *treats response-curve heterogeneity through mixture modelling [hock-*
312 *enberry2017symptom]. None of this comparative machinery has been applied*
313 *to N-of-1 simulation with the Gompertz form as the reference.*

314 **The paper's contribution is a four-family comparison against the standard**
315 **pmsimstats analysis, reported under the ADEMP framework.**

- 316 • Four families are compared: Gompertz, symmetric logistic, hyperbolic-
317 tangent, and piecewise-linear breakpoint.
- 318 • The analysis model is held fixed at the standard pmsimstats linear-mixed
319 specification.
- 320 • Results are reported in the Morris-White-Crowther ADEMP framework
321 adopted as the project-wide standard.

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329 *The present paper's contribution is to import that machinery, specifically:*
330 *a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent,*
331 *piecewise-linear breakpoint) crossed against the standard pmsimstats linear-*
332 *mixed analysis, with results reported in the Morris-White-Crowther [mor-*
333 *ris2019] ADEMP framework already adopted as the project-wide simulation-*
334 *reporting standard.*

2.4 What this paper contributes

Section 2 characterises the pmsimstats Gompertz parameterisation and its relationship to the unified-Richards family.

- The vertical-offset adjustment and calibration parameter sets are documented.
- The unified-Richards reparameterisation provides more interpretable, less correlated parameters.
- This section establishes the reference point for all subsequent sensitivity comparisons.

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In section 2 we characterise the modified Gompertz family as implemented in pmsimstats: the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterisation [tjorve2010richards; tjorve2017] that supplies more interpretable and less correlated parameters.

Section 3 surveys three alternative families at matched marginal effect size for visual and structural comparison with Gompertz.

- Families covered: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Each is calibrated to match the Gompertz reference at the same asymptote and anchor-point response level.
- Shape differences (symmetry, curvature, smoothness) are the primary axes of contrast.

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In section 3 we survey three alternative response-curve families: the symmetric logistic [verhulst1838], the hyperbolic-tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

Sections 4 and 5 report a Monte Carlo sensitivity simulation comparing power, type I error, and bias across the four trajectory families.

- Each family generates trajectories; all are analysed under the fixed standard pmsimstats linear-mixed model.
- Performance measures are reported with conventional Monte Carlo standard errors.
- The design borrows the Benzekry eight-model comparative methodology and the Jagadeesan sloppiness diagnostic.

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In sections 4 and 5 we conduct a Monte Carlo sensitivity-analysis simulation that generates trajectories under each of the four families and analyses them under the standard pmsimstats linear-mixed model. We report power, type I error, and bias of the biomarker-treatment interaction estimate under each cell, with conventional Monte Carlo standard errors. The simulation borrows the comparative methodology of @benzekry2014classical (eight-model tumour-growth comparison) and the sloppiness diagnostic of @jagadeesan2023sloppiness for small-sample identifiability assessment.

Section 6 identifies the trajectory regimes where Gompertz misspecification would be inferentially consequential and recommends sensitivity-grid extensions.

- Consequential regimes include non-saturating growth, biphasic patterns, and breakpoint behaviour.
- Standard N-of-1 inference is materially affected by family choice in these regimes.
- Specific sensitivity-grid extensions are recommended for pmsimstats users.

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In section 6 we identify the trajectory regimes in which Gompertz misspecification would be expected to be consequential: non-saturating growth beyond the trial window, biphasic patterns, and breakpoint behaviour. For these regimes, the standard N-of-1 inference is materially affected by the family choice, and we recommend specific sensitivity-grid extensions that pmsimstats users should run alongside their primary cells.

A companion technical document preserves full mathematical and historical context omitted from the manuscript proper.

- Location: docs/25-gompertz-model-evaluation.md.

- Contains citations, derivations, and historical context that did not survive manuscript editing.
- The manuscript is self-contained; the companion document is supplementary reference material.

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The full mathematical and historical context, including citations that did not survive editing into the manuscript proper, is preserved in the companion technical document at docs/25-gompertz-model-evaluation.md.

3 The modified Gompertz function in pmsimstats

The pmsimstats framework adopts a modified monotone-saturating Gompertz function for each of three latent response components.

- Provenance traces to Gompertz's 1825 mortality law and Winsor's 1932 growth-curve application.
- Three components are parameterised: biological response BR, placebo-belief response PB, and natural-history TV.
- The modified form is monotone-saturating, distinguishing it from the original mortality-curve application.

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The Gompertz curve originates in Gompertz's 1825 law of human mortality [gompertz1825] and was first applied as a general growth curve by @winsor1932gompertz; the pmsimstats framework adopts the modified, monotone-saturating form of that curve. The framework parameterises each of the three latent response components (biological response BR, placebo-belief response PB, and time-variant natural-history component TV) with a modified Gompertz function

$$y(t) = \max r \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)), \quad (1)$$

The three-parameter Gompertz form is asymmetric and monotone-saturating, with the inflection location determined jointly by rate and displacement.

- Parameters are maxr (asymptote), rate (approach speed), and disp (curvature near $t = 0$); a vertical offset ensures $y(0) = 0$.
- Inflection at $t^* = \log(\text{disp})/\text{rate}$; the asymmetry is the key structural distinction from logistic.
- Prazosin/PTSD reference values are $\text{maxr}_{BR} \approx 4$, $\text{rate}_{BR} \approx 0.5$, $\text{disp}_{BR} \approx 4$.

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with vertical-offset adjustment so that $y(0) = 0$. The three parameters are maxr (asymptotic maximum response, matching the on-drug or on-natural-history saturation level), rate (which controls how quickly the response approaches its asymptote), and disp (which controls the curvature near $t = 0$). The Gompertz form is monotone-saturating and asymmetric: the inflection occurs at $t^ = \log(\text{disp})/\text{rate}$, with the rise being relatively slower before the inflection and faster after. The parameterisation used at the prazosin/PTSD reference cell sets $\text{maxr}_{BR} \approx 4$ nightmares per week, $\text{rate}_{BR} \approx 0.5$ per week, and $\text{disp}_{BR} \approx 4$ for biological response; analogous parameter sets for PB and TV are documented in `docs/25-gompertz-model-evaluation.md` and `R/utilities.R`.*

4 Alternative response-curve families

Three alternative families are calibrated to match Gompertz at a common anchor point, isolating shape effects from amplitude effects.

- Families compared: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Calibration constraint: matched asymptote and response level at $t = 5$ weeks via uniroot or closed form.
- Families differ in smoothness, inflection symmetry, and curvature while sharing the same marginal effect size.

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Three alternative families serve as sensitivity-analysis comparators, and in this section we shall describe each in turn before the simulation study proper. Each is parameterised so that the asymptotic maximum and a single-anchor calibration point (the response level at $t = 5$ weeks, calibrated to the Gompertz

family at the same anchor) are matched across families. That is, the families differ in shape envelope (smoothness, symmetry around the inflection, and curvature) while sharing the same marginal effect size at the anchor, a design choice that we discuss further in section 4.

4.1 Symmetric logistic

The logistic function

$$y(t) = \frac{\text{maxr}}{1 + \exp(-\text{rate} \cdot (t - t_0))} - \text{offset} \quad (2)$$

The symmetric logistic rises faster than Gompertz before the inflection and slower after, with symmetric inflection as the principal structural contrast.

- Symmetric around inflection at t_0 ; a vertical offset ensures $y(0) = 0$.
- At matched asymptote and anchor, the logistic envelope differs visibly from Gompertz in pre- and post-inflection curvature.
- Inflection symmetry is the primary qualitative contrast with the asymmetric Gompertz.

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is monotone-saturating and symmetric around its inflection at $t = t_0$. A vertical offset is subtracted so that $y(0) = 0$. At matched maxr and matched response level at the calibration anchor, the logistic family rises faster than Gompertz before the inflection and slower after, producing a visibly different shape envelope while sharing the same asymptote. The symmetric inflection is the principal qualitative contrast with the asymmetric Gompertz curve, and the reader will note that this symmetry property is, in a sense, the logistic family's most theoretically interesting feature for the purposes of the present comparison.

4.2 Hyperbolic-tangent

The hyperbolic-tangent variant

$$y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \quad (3)$$

The hyperbolic-tangent family represents the fast- saturating smooth regime that neither the logistic nor Gompertz can span with standard parameterisations.

- Symmetric and monotone-saturating, but with a steeper pre-asymptote rise than the logistic.

- Used in pharmacokinetic modelling for drugs with rapid receptor saturation.
- Its inclusion spans the fast-saturating smooth regime absent from the other three families.

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is also monotone-saturating and symmetric, but with a qualitatively faster approach to the asymptote than the logistic. The tanh family produces the steepest pre-asymptote rise of the three smooth families, and it has been used in pharmacokinetic modelling for drugs with rapid receptor saturation. We include it here as the representative of the fast-saturating smooth regime that neither the logistic nor the Gompertz family can span with their standard parameterisations.

4.3 Piecewise-linear breakpoint

The piecewise-linear-breakpoint family, formalised in the broken-line regression literature¹,

$$y(t) = \text{maxr} \cdot \min(1, t/t_{\text{bp}}) + \text{post_slope} \cdot \max(0, t - t_{\text{bp}}) \quad (4)$$

The piecewise-linear breakpoint is the canonical non- smooth alternative and provides the strongest test of the framework's sensitivity to trajectory non-smoothness.

- A linear ramp to breakpoint t_{bp} then constant or small post-breakpoint slope.
- The breakpoint is the sole non-differentiable feature across the four families.
- Whether the Wald test detects the non-smooth feature is the central empirical question the simulation addresses.

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is the canonical non-smooth alternative: a linear ramp up to a breakpoint t_{bp} , then constant (when $\text{post_slope} = 0$) or with a small post-breakpoint slope. The breakpoint is the only non-differentiable feature across the four families and is the principal qualitative contrast with the three smooth families. This family is the strongest available test of the framework's sensitivity to

585 *non-smoothness in the trajectory. But how different is the piecewise-linear*
586 *breakpoint from the smooth families, from the perspective of the Wald test that*
587 *the analysis model applies? That question is precisely what the simulation in*
588 *section 4 is designed to answer.*

589 5 Sensitivity-analysis simulation design

590 **The primary estimand is the bm:Dbc interaction coefficient; performance**
591 **measures are power, type I error, bias, and empirical Monte Carlo SE.**

- 592 • Analysis model: $Sx \sim bm + t + Dbc + bm : Dbc + (1 \mid ptID)$ with
593 $corCAR1(form = \sim t \mid ptID)$.
- 594 • Primary outputs: power at $\alpha = 0.05$ and type I error under $\beta_{bm:D} = 0$.
- 595 • Secondary outputs: bias and empirical MC SE of $\hat{\beta}_{bm:D}$; all proportions
596 reported with binomial MCSE.

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604 *The primary estimand is the biomarker-by-treatment interaction coefficient*
605 *$\hat{\beta}_{bm:D}$ from the linear mixed-effects analysis model $Sx \sim bm + t + Dbc + bm :$*
606 *$Dbc + (1 \mid ptID)$ with $corCAR1(form = \sim t \mid ptID)$. Primary inferential out-*
607 *puts are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under*
608 *$\beta_{bm:D} = 0$. Secondary estimands are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated*
609 *true effect and the empirical Monte Carlo standard error of $\hat{\beta}_{bm:D}$ across repli-*
610 *cates. Performance measures are reported with binomial-proportion Monte*
611 *Carlo standard errors for proportions and the standard Monte Carlo SE of*
612 *the within-replicate mean for continuous quantities.*

613 **The ADEMP design is pre-registered and fixes all four DGP families, the**
614 **calibration sub-study, and the full factorial before production runs.**

- 615 • Pre-registration location: analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md.
- 616 • Study 0 calibrates all four families to a common marginal effect size; Study
617 1 is the 144-cell factorial; Study 2 is the sloppiness eigenvalue diagnostic.
- 618 • Pre-registration prevents post-hoc specification of performance measures
619 or DGP choices.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration`, which fixes the four DGP families (modified Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study (Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

The present submission uses a focused 16-cell factorial scoped to answer whether trajectory family matters for the `bm:Dbc` Wald test at MCSE 0.013–0.016.

- Design: 4 families $\times$ 2 architectures $\times$ 2 effect sizes; prazosin-calibrated OL+BDC, $N = 35$, $t_{1/2} = 0$.
- Carryover is absent to isolate trajectory shape from carryover-mediated correlation decay.
- The 144-cell extended grid is deferred to a follow-up; the 16-cell factorial addresses the central question at adequate MCSE resolution.

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For the present submission the simulation is scoped to a 16-cell focused factorial: four trajectory families crossed with two DGP architectures (mean moderation and MVN differential correlation, following the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`) crossed with two biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-calibrated Hybrid OL+BDC, $N = 35$, $t_{1/2} = 0$, with carryover set to absent in order to isolate the trajectory-shape effect from carryover-mediated correlation decay. Each cell is run at $n_{\text{reps}} = 1,000$ replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is deferred to a follow-up; the focused 16-cell factorial answers the central scientific question (does trajectory family choice matter for the `bm:Dbc` Wald test?) at MCSE 0.013–0.016 per cell.

All four families are matched at $t = 5$ weeks with the asymptotic maximum held constant; calibrated parameters are recorded for reproducibility.

- Anchor calibration: each family's response at $t = 5$ matched to Gompertz via uniroot or closed form.
- Asymptotic maximum held constant across all four families.
- Calibrated parameters recorded in `01-architecture-a-trajectory-sweep.R` and read back by manuscript chunks.

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Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family's response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via uniroot (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver analysis/scripts/gompertz-evaluation/01-architecture-a-trajectory-sweep.R and read back by the manuscript chunks for transparency.

6 Results

The 16-cell simulation completed with 100% convergence in 381 s; cell MCSE on power runs from 0.013 to 0.016.

- 1,000 replicates per cell across 16 cells; 8-core parallel::mclapply.
- Per-replicate output at analysis/data/quick-sim/07-gompertz-replicates.rds; ADEMP summary at 07-gompertz-summary.txt.
- Results are presented in order: power, type I error, bias.

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The simulation programme was executed at 1,000 replicates per cell across the 16-cell trajectory-family-by-architecture-by-effect-size factorial in 381 s using 8-core parallel::mclapply. Convergence was 100% across all 16,000 fits. Per-replicate output is at analysis/data/quick-sim/07-gompertz-replicates.rds; the Morris ADEMP cell-level summary is at analysis/data/quick-sim/07-gompertz-summary.txt. Cell MCSE on power runs from 0.013 to 0.016 across the alternative cells and from 0.005 to 0.006 on the conservative null cells. We begin the results section with power, then turn to type I error, and finally consider bias of the interaction estimate.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$ under both architectures, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	76.1 $\pm$ 1.4	78.1 $\pm$ 1.3
Logistic	77.1 $\pm$ 1.3	80.6 $\pm$ 1.3
Hyperbolic tangent	76.9 $\pm$ 1.3	78.0 $\pm$ 1.3

Family	Architecture A	Architecture B
Piecewise-linear breakpoint	77.8 ± 1.3	77.9 ± 1.3

Power is essentially invariant to trajectory family under both architectures; the pre-registered piecewise-linear hypothesis is not supported.

- Architecture A: four families within a 0.017 spread, well inside two cell-MCSE.
- Architecture B: spread 0.027; logistic at 0.806 is marginal (≈ 2 MCSE above others) and not reproduced under Architecture A.
- Piecewise-linear breakpoint is indistinguishable from Gompertz and hyperbolic-tangent under both architectures.

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We note that under Architecture A (mean moderation), the four families cluster within a 0.017 spread (1.7 percentage points), well inside two cell-MCSE of sampling noise. Under Architecture B (MVN differential correlation), the spread is 0.027; the logistic family at 0.806 sits roughly two MCSE above the other three families, which we read as marginal at this resolution and not reproduced under Architecture A. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported: at MCSE 0.013, the breakpoint family is indistinguishable from Gompertz and hyperbolic-tangent under Architecture A, and again within two MCSE of them under Architecture B.

The earlier 200-replicate prototype result was statistically consistent with the production run but was under-powered to distinguish the apparent gap from noise.

- Prototype: Gompertz 0.765, logistic 0.805 under Architecture B; apparent gap 0.040, roughly 1.5 prototype-MCSE.
- Production: gap settles to 0.025, within two production-MCSE of zero.
- The prototype was not wrong; it lacked power to distinguish signal from noise at its replicate count.

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The 200-replicate prototype reported in the earlier draft of this manuscript

(Gompertz 0.765, logistic 0.805 under Architecture B alone) is statistically consistent with the present production-replicate results: the prototype's apparent 0.040 Gompertz-logistic gap was roughly 1.5 prototype-MCSE, and at the 1,000-replicate resolution it settles to 0.025, itself within two production-MCSE of zero. That is, the prototype result was not wrong, but it was under-powered to distinguish signal from noise at that resolution.

6.2 Type I error across response-curve families

Type-I error at $c_{bm} = 0$, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	2.6 ± 0.5	2.7 ± 0.5
Logistic	3.2 ± 0.6	2.9 ± 0.5
Hyperbolic tangent	2.9 ± 0.5	2.5 ± 0.5
Piecewise-linear breakpoint	2.8 ± 0.5	3.6 ± 0.6

Type I error is slightly conservative and uniform across all four families, consistent with corCAR1 small-sample degree-of-freedom accounting.

- All eight null cells fall in $[0.025, 0.036]$; no family-specific inflation is observed.
- The conservatism is shared across families, implicating the corCAR1 d.f. mechanism rather than trajectory shape.
- This pattern is consistent with prior pmsimstats runs at reference parameters and $N = 35$.

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All eight null cells produce empirical type I error in $[0.025, 0.036]$, slightly conservative relative to the nominal 0.05 and consistent with the corCAR1 small-sample degree-of-freedom accounting at the prazosin-calibrated $N = 35$ and 8-timepoint short serial series. There is no family-specific inflation. The conservatism is shared across all four families, supporting the interpretation that it reflects the corCAR1 d.f. mechanism rather than a trajectory-shape artefact. We note that this pattern of slight conservatism has appeared consistently across pmsimstats simulation runs at the reference parameters, and it is relatively well-understood as a consequence of the corCAR1 variance-component estimation at small N .

6.3 Bias of the biomarker-treatment interaction estimate

Bias of the interaction estimate is negligible and uniform across families; within-replicate model SE matches empirical SE across all cells.

- Mean $\hat{\beta}_{bm:D_{bc}}$ varies from -0.230 to -0.239 across the eight alternative cells, a 0.009 spread within Monte Carlo noise.
- Empirical SE is essentially constant across families (0.074 to 0.082).
- Model SE matches empirical SE, confirming well-calibrated nominal-95% Wald coverage.

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Across the eight alternative cells the mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ varies over a narrow range (roughly -0.230 to -0.239), a cell-to-cell spread of about 0.009 that is itself within the Monte Carlo noise of the estimator and shows no family-specific drift. The empirical SE of $\hat{\beta}_{bm:D_{bc}}$ is essentially constant across families (SD in the range 0.074 to 0.082 across cells) and roughly equal between architectures. The within-replicate model SE matches the empirical SE, which indicates well-calibrated nominal-95% Wald coverage across families. In sum, the bias results are unremarkable in the best sense: there is no trajectory-family signal here that the power and type I error results might have concealed.

7 Discussion

7.1 A negative finding, cleanly resolved

Trajectory family choice has no detectable effect on the bm:D_{bc} Wald test at the prazosin reference cell under either DGP architecture.

- Four families spanning asymmetric, symmetric-smooth, and non-smooth templates all produce statistically indistinguishable power and type I error.
- The finding is a null result at one reference cell and should not be overstated.
- Conditions under which the result might not hold are discussed in section 6.3.

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We report what is, in essence, a negative finding: at the prazosin-calibrated reference cell (Hybrid OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), trajectory family choice has no detectable effect on the bm:D_{bc} Wald test under either DGP architecture. The four families span the

qualitative range of plausible templates for monotone-saturating treatment-response trajectories: asymmetric (Gompertz), symmetric smooth (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint). All four produce statistically indistinguishable power and type I error at MCSE 0.013 to 0.016 per cell. We would not wish to overstate this finding; it is a null result at one reference cell, and the conditions under which the result might not hold are discussed in section 6.3 below.

The pre-registered piecewise-linear hypothesis is not supported; the breakpoint feature is invisible to the Wald test in the explored regime.

- The non-differentiable breakpoint carries no detectable power cost at $N = 35$ and $c_{bm} = 0.45$.
- Smoothness assumptions in the trajectory family carry no inferential cost in the regime evaluated.
- This null result is methodologically informative: it implies the Wald test is robust to the non-smooth feature.

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The pre-registered hypothesis that the piecewise-linear breakpoint family, with its non-differentiable feature, would behave qualitatively differently from the smooth families is not supported. The breakpoint feature is invisible to the linear-mixed Wald test at the explored sample size and effect size. This is a methodologically valuable null result: if the breakpoint feature were detectable through power loss, the manuscript would have reported a meaningful sensitivity to non-smoothness. Its absence implies that smoothness assumptions in the trajectory family carry no inferential cost in the regime evaluated.

7.2 Why trajectory family does not matter at this resolution

Three mechanisms jointly account for the trajectory- family null finding.

- Calibration absorbs shape differences at the common anchor, isolating shape from amplitude effects.
- The bm:Dbc Wald test integrates over time, averaging local shape differences within the predictor structure.
- Under Architecture B, biomarker-treatment information lives in the covariance structure, upstream of and independent from trajectory shape.

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We see three mechanisms that, taken together, account for the negative finding, and we shall work through each in turn.

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Matched-anchor calibration ensures that residual family differences are in shape only, not in amplitude, and those shape differences are small relative to between-replicate variation.

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- Families matched at asymptote and $t = 5$ anchor.

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- Within-trajectory shape variation is small relative to the between-replicate variance of the `bm:Dbc` estimator.

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- The calibration design isolates shape effects from amplitude effects by construction.

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The first mechanism is that calibration absorbs the differences. The four families are matched at the asymptotic maximum and at a single anchor point ($t = 5$ weeks). At matched marginal effect size, the within-trajectory variation in shape is small relative to the between-replicate variation in the `bm:Dbc` estimator. The calibration constraint is intentional in the simulation design: it isolates trajectory-shape effects from amplitude effects, so that any family-specific power differences we observe can be attributed to shape alone.

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The `bm:Dbc` Wald test averages over on-drug timepoints, smoothing out local trajectory-shape differences that would be visible in point-in-time comparisons.

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- The continuous D_{bc} predictor integrates trajectory contributions across all on-drug timepoints.
- Local shape differences are averaged within the integration and do not accumulate to detectable power differences.
- This temporal averaging appears to be the dominant mechanism behind the near-identical power.

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The second mechanism is that the `bm:Dbc` Wald test averages over time. The continuous D_{bc} predictor multiplied by the biomarker integrates the trajectory's contribution across all on-drug timepoints, with weights determined

918 by the predictor's exponential-decay structure. Local trajectory-shape differ-
919 ences (where one family rises faster than another between timepoints) are
920 averaged within the integration, and the Wald test is therefore relatively in-
921 sensitive to local-shape features. It appears that this averaging property is the
922 dominant reason for the near-identical power across families.

923 **Under Architecture B, the biomarker-treatment information channel is**
924 **in the covariance structure and is independent of trajectory shape by**
925 **construction.**

- 926 • Architecture B encodes information in $\partial\text{Cor}(BM, BR)/\partial D_{bc}$, not in the
927 trajectory mean.
- 928 • This channel is independent of the trajectory family that BR follows.
- 929 • Family choice cannot modulate Architecture B's information by construc-
930 tion; invariance is expected.

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938 *The third mechanism applies specifically to Architecture B: the covariance*
939 *moderation is upstream of trajectory shape. Under MVN differen-*
940 *tial correlation, the biomarker-by-treatment information lives in the*
941 *$\partial\text{Cor}(BM, BR)/\partial D_{bc}$ structure, which is independent of the trajectory*
942 *family that BR follows. Family choice cannot affect Architecture B's*
943 *information channel by construction.*

944 **Trajectory family is inferentially inert at this resolution; the modGompertz**
945 **default is robust by the combined action of the three mechanisms.**

- 946 • Calibration, temporal averaging, and covariance-upstream encoding jointly
947 suppress family-specific power variation.
- 948 • The modGompertz default is robust at the reference parameters under both
949 architectures.
- 950 • The following section identifies conditions where this robustness would be
951 expected to break down.

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959 *The combined effect is that trajectory family choice is, at this resolution,*
960 *an inferentially inert axis of the simulation design, and the framework's*
961 *modGompertz default is in this sense robust by construction. We turn next to*
962 *the conditions under which that robustness would be expected to break down.*

7.3 Conditions that would change the answer

Four conditions outside the present 16-cell factorial would plausibly produce trajectory-family-dependent power.

- Smaller N , different c_{bm} , non-zero carryover ($t_{1/2} > 0$), and non-monotone or biphasic shape envelopes.
- None of these conditions is exercised in the present focused factorial.
- Each is a natural axis for the 144-cell follow-up grid.

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Four conditions, none exercised in the present 16-cell factorial, would plausibly produce trajectory-family-dependent power, and each merits a brief comment.

Smaller sample size might amplify local-shape sensitivity by concentrating temporal information at fewer timepoints.

- At $N = 35$, the test integrates over a moderately large within-subject sample.
- With fewer timepoints, local shape differences between families would be more visible rather than averaged.
- This condition is the most proximate candidate for a departure from the null result.

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First, smaller sample size. At $N = 35$ the $bm:Dbc$ test integrates over a moderately large within-subject sample; smaller N might amplify the local-shape sensitivity by concentrating the temporal information at fewer timepoints where the families differ more visibly.

At smaller c_{bm} near the 50% power boundary, family differences might surface via different finite-sample sampling distributions.

- The $c_{bm} = 0.45$ cell sits in a moderate-power regime with comparable signal-to-noise across families.
- Pushing to $c_{bm} = 0.30$ (power near 0.5) may reveal family-specific finite-sample sampling differences.
- The tight clustering at 0.45 may be a local feature of the effect-size space.

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1013 *Second, larger or smaller c_{bm} . The $c_{bm} = 0.45$ reference cell sits in a power*
1014 *regime where families have comparable signal-to-noise. Pushing c_{bm} to 0.30*
1015 *(where power lies near 0.5) might surface family differences via different finite-*
1016 *sample sampling distributions, and it is possible that the tight clustering we*
1017 *observe is a feature of this particular region of the effect-size space.*

1018 **Non-zero carryover introduces an interaction between trajectory shape**
1019 **and the off-drug exponential decay that is suppressed in the present $t_{1/2} =$**
1020 **0 slice.**

- 1021 • With $t_{1/2} > 0$, integration weights interact with the off-drug decay, poten-
1022 tially differentiating families.
- 1023 • The present simulation sets $t_{1/2} = 0$ to isolate trajectory shape from car-
1024 ryover effects.
- 1025 • This interaction is the mechanistic pathway by which carryover could make
1026 trajectory family consequential.

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1034 *Third, carryover-induced correlation decay. With $t_{1/2} > 0$, the $bm:Dbc$ -*
1035 *mediated information channel is eroded differently for different trajectory*
1036 *shapes, because the integration weights interact with the off-drug exponential*
1037 *decay. The $t_{1/2} = 0$ slice exercised here suppresses this interaction entirely.*

1038 **Non-monotone or biphasic trajectory families would produce genuinely dif-**
1039 **ferent $bm:Dbc$ information content to which the Wald test would almost**
1040 **certainly respond.**

- 1041 • All four present families share a monotone-saturating envelope; the null
1042 result is conditional on this property.
- 1043 • Biphasic or oscillating families have qualitatively different information struc-
1044 ture.
- 1045 • Extension to non-monotone families is outside the scope of the present
1046 study.

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Fourth, non-monotone or biphasic shape envelopes. The four families we have examined share a monotone-saturating envelope. Families that peak and decay (biphasic) or oscillate would produce genuinely different trajectories with different $bm:Dbc$ information content, and the Wald test would almost certainly respond to those differences.

The 144-cell extended grid is the canonical follow-up that would close the trajectory-family question at publication- adequate rigour.

- The grid crosses N , c_{bm} , $t_{1/2}$, and family.
- Each of the four conditions above maps to a dimension of this grid.
- The present focused factorial is a necessary precursor; the extended grid is the production-grade completion.

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Each condition is the natural axis of a follow-up sensitivity analysis. The 144-cell grid anticipated in the abstract ($N \times c_{bm} \times t_{1/2} \times \text{family}$) is the canonical follow-up, and we recommend it as the production-grade extension that would close out the trajectory-family question at a level of rigour adequate for the methodological literature.

7.4 The prazosin/PTSD reference cell

For the prazosin-PTSD application, trajectory family is the least consequential sensitivity axis; attention should be directed to carryover and DGP-architecture.

- Sensitivity to modGompertz is small relative to carryover specification and DGP-architecture sensitivity.
- The modGompertz choice can be defended without reservation at the reference parameters.
- Sensitivity-analysis resources are better spent on carryover and DGP-architecture axes with demonstrably larger inferential consequences.

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1094 *For the motivating prazosin-PTSD application that drives the pmsim-*
1095 *stats research programme, the present result implies that sensitivity*
1096 *to the modGompertz assumption is small relative to sensitivity to the*
1097 *carryover specification (the subject of the companion manuscript at*
1098 *analysis/report/02-carryover-sensitivity/) and to the DGP architec-*
1099 *ture (the subject of analysis/report/01-dgp-mean-moderation-vs-mvn/).*
1100 *That is, of the three principal axes of sensitivity uncertainty in the pmsim-*
1101 *stats simulation framework, trajectory family choice is, at the reference*
1102 *parameters, the least consequential by a substantial margin. A trial*
1103 *designer using the pmsimstats framework can defend the modGompertz*
1104 *choice without reservation at the reference parameters, and we would*
1105 *suggest that sensitivity-analysis attention be redirected to the carryover and*
1106 *DGP-architecture axes, where the inferential consequences are demonstrably*
1107 *larger.*

1108 8 Conclusions

1109 **The modified Gompertz is a defensible template for monotone-saturating**
1110 **N-of-1 trajectories; four-family comparison produces indistinguishable**
1111 **power and type I error.**

- 1112 • Power and type I error are statistically indistinguishable across all four
1113 families under both architectures at MCSE 0.013–0.016.
- 1114 • The piecewise-linear breakpoint hypothesis is not supported.
- 1115 • The Gompertz default is defensible at the reference parameters.

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1123 *First, the modified Gompertz function is a defensible parametric template*
1124 *for monotone-saturating treatment-response trajectories in N-of-1 simulation.*
1125 *At the prazosin/PTSD reference cell with matched-anchor calibration, the*
1126 *four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent,*
1127 *piecewise-linear breakpoint) produces statistically indistinguishable power and*
1128 *type I error under both DGP architectures at MCSE 0.013 to 0.016 per cell.*
1129 *The pre-registered hypothesis that the piecewise-linear breakpoint family would*
1130 *behave qualitatively differently from the smooth families is not supported.*

1131 **The null result implies the Wald test is robust to principal trajectory-**
1132 **family axes in the evaluated regime; boundary conditions are deferred to**
1133 **the 144-cell follow-up.**

- 1134 • Robustness holds across asymmetric, symmetric-smooth, and non-smooth
1135 families in the evaluated regime.

- At what sample sizes, effect sizes, or carryover levels robustness breaks down is an open question.
- Sensitivity-analysis priority should go to carryover and DGP-architecture axes with larger demonstrated consequences.

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Second, the negative finding is methodologically informative. It implies that the `bm:Dcb` Wald test is robust to the principal qualitative axes of trajectory-family choice in the regime evaluated. Unfortunately, we cannot yet establish at what sample sizes, effect sizes, or carryover levels that robustness would begin to break down; that question is deferred to the 144-cell follow-up. Trial designers using the `pmsimstats` framework can accept the `modGompertz` default without reservation at the reference parameters and direct sensitivity-analysis attention to the carryover and DGP-architecture axes that have demonstrably larger inferential consequences.

The 144-cell extended grid is the work needed to close the trajectory-family question at publication-adequate rigour.

- The present 16-cell factorial addresses the central question with statistical clarity.
- The extended grid would map boundary conditions at which trajectory family begins to matter.
- This follow-up is recommended as the production-grade completion of the trajectory-family evaluation.

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Third, the 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is the natural follow-up, and we regard it as the work that would be needed to close the trajectory-family question at a level of rigour adequate for the methodological literature. The present focused 16-cell factorial addresses the central scientific question with statistical clarity; the extended grid would map the boundary conditions at which trajectory-family choice begins to matter.

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

Data and code are openly available in the pmsimstats-ng repository under versioned standard paths.

- Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are at `analysis/data/`.
- Simulation drivers are at `analysis/scripts/`.
- Exact manuscript paths are listed in the Reproducibility section.

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

The manuscript is fully reproducible inside the zzcollab Docker container with the renv.lock package set.

- Container image hash in Dockerfile; package set in `renv.lock`.
- Principal packages: `nlme`, `parallel`, `data.table`, `furrr/purrr`, `rmarkdown/bookdown`.
- Rendered with R 4.4.x.

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This manuscript was rendered with R 4.4.x inside the project's zcollab Docker container (image hash recorded in Dockerfile; package set captured in renv.lock). Principal packages used by the simulation pipeline are nlme (continuous-time linear-mixed analysis with corCAR1 residual correlation), parallel (8-core mclapply for parameter-grid sweeps), data.table (the original-extended simulation collection), furrr and purrr (the tidyverse simulation collection), and rmarkdown plus bookdown for the manuscript build.

Seeds, checkpoints, and driver paths are fully specified for exact replication of the simulation results.

- Master seed `set.seed(20260507)`; per-replicate seeds via `parallel::mc.set.seed` with `+ rep_idx` offsets.
- Outputs checkpointed at `analysis/data/quick-sim/07-gompertz-replicates.rds`; ADEMP summaries at `07-gompertz-summary.txt`.
- Pre-registered ADEMP plan at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`.

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Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/07-gompertz-replicates.rds`; Morris ADEMP cell-level summaries are at the companion `07-gompertz-summary.txt`. Driver scripts are under `analysis/scripts/gompertz-evaluation/`. The pre-registered ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/07-gompertz-evaluation/`.

13 References

1. Muggeo VMR. Estimating regression models with unknown break-points. *Statistics in Medicine*. 2003;22(19):3055-3071. doi:10.1002/sim.1545

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